



Potential of ethno-veterinary medicine in animal health care practices in the South Pacific Island countries: a review

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Abstract

The South Pacific island countries (SPIC) are among the most vulnerable countries in the world to economic shocks, climate change and natural disasters. Agriculture including livestock production makes important contribution to socio-economic development in the region. However, the livestock sector is marred by many livestock diseases, which are crippling the economies as well as threatening the food security of the island nations, most of which have high dependence on imports. Limited veterinary medical facilities, high cost of conventional medicine and remoteness of some rural farming communities make it difficult for farmers to treat their diseased animals. Nevertheless, the island nations are endowed with vast reserves of medicinal plants in diverse ecosystems including mangroves, rainforests and tropical dry forests. Despite the ubiquity of traditional medicine systems, most of the plant species have been used to treat human ailments in the SPIC and very few farmers utilise herbal medicine to treat their animals. This review highlights the common diseases affecting livestock production in the SPIC, threats to ethno-veterinary knowledge, and potential plant species and their efficacies to treat animal diseases and parasites. It also discusses the common methods of preparation of veterinary herbal medicine and possible ways of making the medicine available in the formal and informal markets in the region. It is envisaged that the review will stimulate further ethno-veterinary research among livestock disease management practices in the SPIC.

Keywords Medicinal plants · Livestock diseases · Parasites · Indigenous knowledge · South Pacific Island countries

Introduction

In the South Pacific Island countries (SPIC), most people wholly rely on fishing and agricultural activities like growing crops and rearing of livestock for sustenance (Reddy 2007). Livestock production is based on farm animals such as cattle, sheep, goats, pigs and poultry, which are a ready source of protein (milk and meat) and can be sold to generate cash for meeting other household needs. Most of the animals are

managed under extensive production system and very few commercial farms exist in the SPIC. Nevertheless, the importance of livestock in the region is widely recognised, especially among the peasant and/or small-scale farmers living below the poverty datum line.

The climate of SPIC is mostly tropical and supports a wide variety of plant species. The rainforests fill the valleys of the high islands whereas low islands are restricted to a few hardy and drought-resistant plant species (Moorhead 2011). Most of the plant resources in the region are under-utilised. If fully exploited, animal production costs could be significantly reduced. Some of the potential uses of plant resources from the islands include use as feed and traditional medicine for livestock. With regards to traditional medicine, a number of plant species have been identified to treat a wide range of human ailments. Use of plants as traditional medicine for treating human ailments has been reported in Vanuatu (Bradacs et al. 2011), Samoa (Harrington 2001), Cook Islands, Solomon Islands, Tonga, Tahiti, Tuvalu, Niue (WHO 1998), Nauru (WHO 2002) and Papua New Guinea (WHO 2009; Jorim et al. 2012). Likewise,

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traditional medicines from plants have been used by almost all ethnic groups in Fiji (Singh 1986). Knowledge on the use of traditional medicine from plants is, however, restricted to elderly people, and with increasing acculturation and rural-urban migration, there is a high risk of its erosion among the new generation (Furusawa 2006). As in humans, traditional medicine can also be used to treat animal diseases.

Animal health is a major constraint to livestock production in the SPIC. However, little, if any, information has been documented on the use of traditional medicine for treating animal ailments. There are a number of diseases causing major production losses and threats to human health. Some of the common diseases and parasites in the south pacific are bovine tuberculosis, leptospirosis, brucellosis, tick borne diseases and gastrointestinal parasites (Brioudes et al. 2015; Borja et al. 2018). Proliferation of most of the diseases is a result of conducive hot and humid conditions of the tropical climate. Moreover, changes in the climate may lead to infectious diseases spreading to regions where they are not commonly reported (De La Rocque et al. 2008). Animal disease management in the region is partly achieved through the use of conventional veterinary medicine, forming quarantine camps and controlling movement of animals to and from affected areas. However, due to the remoteness of rural farming communities in many islands and high cost of the conventional drugs, it becomes impossible to control some of the animal diseases. Therefore, the use of herbal plants could be a substitute for the conventional drugs. The use of herbal plants has many advantages such as low cost, readily available and no drug residues are left in milk and meat products (Dhama et al. 2015). Furthermore, the use of herbal plants could be sustainable and adaptable to the rural farming communities in SPIC.

In Africa and other developing countries, use of ethno-veterinary medicine is proving to be successful. Around 80% of the marginalised people in the world depend on traditional medicine for controlling and treating various diseases affecting human beings and animals (Mahomoodally 2013). There is a vast body of literature in their region on the different plants to use for treating their livestock. Hardly do you find farmers in SPIC using traditional/ethno-veterinary medicine for their livestock or literature on different plant species used to treat livestock diseases. The aim of this review is to collate information on potential plant species that can be used to treat common livestock diseases in the south pacific region.

Difficulties and challenges of livestock health management in the SPIC

The SPIC have a number of endemic diseases that cause production losses. Figure 1 illustrates some of the prioritised livestock diseases in Pacific countries. The diseases are ranked as per surveys on animal disease perception of farmers in the Pacific region (Brioudes et al. 2015). Livestock mortalities,

as a result of the diseases, leave farmers with limited options for survival. In Fiji, for example, bovine tuberculosis and brucellosis have led to decreased production and opportunity for local trade due to high morbidity and early culling of potentially productive stock (Borja et al. 2018). The country, therefore, relies on imports from other countries.

It is noteworthy to mention the devastating effects of climate change on agricultural activities in the world. Livestock health management in the SPIC has also not been spared by the vagaries of nature (SPC 2011). The rise in temperature and change in rainfall patterns have a substantial effect on the encumbrance of infectious diseases that are transmitted by insect vectors and through contaminated water. Studies have shown the infective stage of trematodes (flukes), cercariae, to be positively correlated with temperature (Lo and Lee 1996; Umadevi and Madhavi 1997; Mouritsen et al. 2018). In 2012, leptospirosis outbreak was reported in Fiji after two floods, which caused 576 reported cases and 7% cases of fatality in humans (Togami et al. 2018). It is believed that leptospirosis is spread by rodents and livestock species like cattle, sheep and pigs. Infectious diseases are also spreading to regions where they are not commonly encountered following cyclones and floods. The prevalence of diseases is aggravated by the unavailability of conventional medicine and veterinary personnel in remote and far-flung areas and/or islands. Since most of the Pacific islands have abundant plant resources, their potential utilization as ethno-veterinary medicine may reduce the likelihood of livestock mortality due to infectious diseases. This calls for the need to identify plant species in the SPIC with potential to be used as traditional medicine to treat diseased animals.

Evolution and role of medicinal plants in animal health care

It is believed that the use of medicinal plants in traditional human society evolved through two different pathways (Pieroni et al. 2006). Zoopharmacognosy is one possible evolutionary pathway for ethno-veterinary medicine. Through studying feeding behavioural patterns in apes (Huffman 2001; Huffman 2003) and ungulate species (Hart and Hart 2018) in the wild, a number of plant species with potential to be used as ethno-veterinary medicine were discovered. The second premise is that ethno-veterinary medicine co-evolved with human folk medicine. There exists a high degree of similarity between traditional medicinal practices in humans and animals in terms of the resources and modes of administration (McCorkle and Martin 1998; Souto et al. 2011; Megersa et al. 2013). This implies that plants used for treatment and control of diseases and parasites in humans could have similar uses in animals.

Knowledge of medicinal plants for both livestock and humans is an important asset for the livelihood of the indigent people in the developing countries of the world. The dependence on medicinal plants is higher, where pharmaceutical

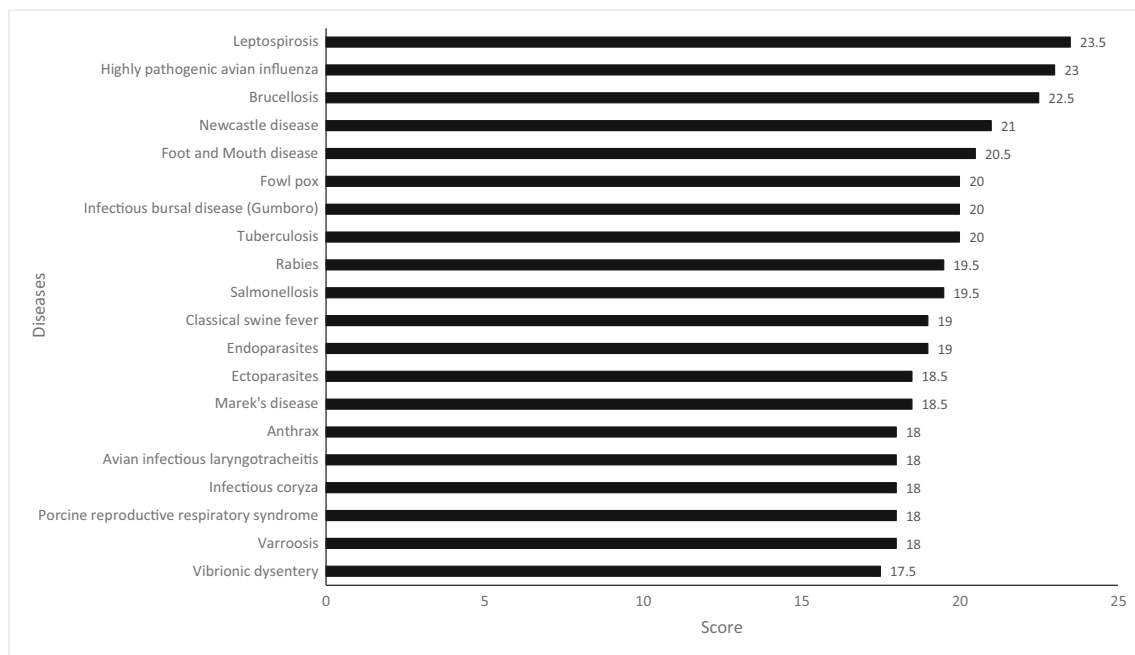


Fig. 1 List of prioritised livestock diseases in the Pacific region ranked by score (Brioude et al. 2015)

drugs are imported and many times inaccessible due to their high costs (Minja 1994; Rajpar et al. 2011). Similarly, in most of the remote and far-flung areas and/or islands where animal health facilities are often very limited and involve travelling long distances to nearest dispensaries, the use of available plant resources as medicine becomes important. However, despite the availability and diversity of bio-resources, the utilisation of phytomedicine for the treatment of animal diseases in the South Pacific is very low. Table 1 shows some of the potential medicinal plants which could be used for treatment of livestock diseases prevalent in the SPIC. These medicinal plant species are also found in Fiji and other South Pacific Islands, and therefore, their utilisation could be possibly explored in ethno-veterinary practices in the region.

Traditional methods of preparing and applying ethno-veterinary medicine

Medicinal plants can be used for preparing herbal remedies through different methods, depending on the plant part utilised and animal condition. Some of the common traditional methods used for preparing plants for medicinal uses include crushing (to extract sap or form a powder), chewing and decoction (Megersa et al. 2013; Banerjee et al. 2015; Araya et al. 2015; Teklay 2015; Kebebew and Mohamed 2017). These methods are some of the practical ways indigent people in SPIC can use when preparing and applying medicines to animals. Other method like homogenisation to extract saps is expensive and also requires technical expertise. However, careful selection of species, plant part and mode of preparation and administration of the herbal medicine play a very critical

role in traditional medicine (Upirety et al. 2012). Therefore, initiatives entailing the use of plant resources for treating livestock diseases in the Pacific would be applicable only after thorough testing and under the supervision of knowledgeable people. Figure 2 is a summary of the common traditional methods used for preparing and applying herbal medicine to animals.

Crushing can be used for both succulent plants to extract the sap and dried plant parts to form a powder. Crushing of plants is normally done by using simple hand tools like pestle and mortar. When succulent plants or plant parts are crushed to extract the sap, it can be applied topically on open wounds to prevent secondary infections and can also be mixed with water for drinking (Mwale et al. 2005). Crushing can also be used for dried plant parts like leaves, stems, bark and roots. Drying is normally done for preservation and crushing done to ensure faster solubility (Abera 2014). Dried medicinal plants in their powder form may be dissolved in solvents such as water which is generally preferred because most active constituents in plants are water soluble (Alade et al. 2015). Thereafter, the herbal preparations can be administered orally to animals. Some of the plant species that are generally crushed to prepare herbal remedies are *Aloe megalacantha*, *Allium sativum* (Araya et al. 2015), *Colocasia esculenta* (Megersa et al. 2013), *Ocimum canum*, *Zingiber officinale* (Panda and Mishra 2016) and *Maesa lanceolata* (Kipkore et al. 2014).

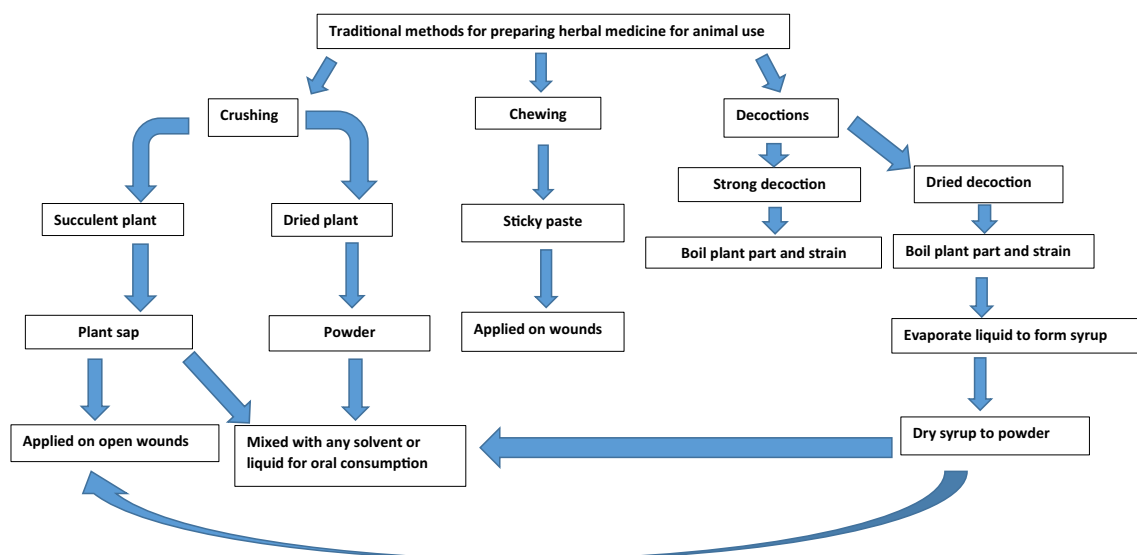
Chewing is another traditional method for preparing herbal medicine. This method is popular in human medicine, especially in tropical regions like the Amazon basin and some Pacific Island nations (Harrington and Scotese 1996;

Table 1 (continued)

Disease	Plant species	Habit	Part used	Country of reported use	Reference
Infectious coryza	<i>Moringa oleifera</i>	Tree	Leaves	Africa	Ogni et al. (2016)
	<i>Capsicum annum</i>	Shrub	Fruits	Africa	Ogni et al. (2016)
Newcastle disease	<i>Aloe vera</i>	Herb	Leaves	Africa	Mwale et al. (2005), Okitoi et al. (2007), Lagu and Kayanja (2010)
	<i>Capsicum annum</i>	Shrub	Seeds	Africa	Guèye (1999), Guèye (2002), Lagu and Kayanja (2010)
	<i>Khaya senegalensis</i>	Tree	Bark	Africa	Guèye (1999), Guèye (2002)
	<i>Mangifera indica</i>	Tree	Bark	Africa	Guèye (2002)
	<i>Allium sativum</i>	Herb	Bulb	–	Sule et al. (2018)
	<i>Amaranthus spinosus</i>	Herb	Seeds	Africa	Pagadala et al. (2015)
Rabies	<i>Datura stramonium</i>	Herb	Leaves	Africa	Admasu and Mekonnen (2014)
Salmonellosis	<i>Moringa oleifera</i>	Tree	Leaves	Africa	Dougnon et al. (2017)
	<i>Ocimum gratissimum</i>	Herb	Leaves	Africa	Dougnon et al. (2017)
	<i>Manihot esculenta</i>	Shrub	Leaves	Africa	Dougnon et al. (2017)
	<i>Carica papaya</i>	Tree	Leaves, seeds	Africa	Dougnon et al. (2017)

Odonne et al. 2013). In humans instead of crushing using pestle and mortar, chewing is used to extract plant juices/sap. Unlike humans, it is difficult for farmers to force animals to chew medicinal plants particularly if the animal is not familiar with the plant and it has to be administered orally. In such cases, humans may chew the plant part in the mouth to form a sticky paste, which can thereafter be applied on the wound of an animal for healing and to stop bleeding. Some plants used for treating animal diseases through chewing are *Rumex nepalensis*, *Cucumis ficifolius* (Megersa et al. 2013), *Achyranthes aspera* (Araya et al. 2015) and *Nicotiana tabacum* (Banerjee et al. 2015).

The aqueous preparations of plant parts are known as decoctions (Nafiu et al. 2017). They are prepared by breaking the plant into small pieces before soaking in water and boiling for a specified time. There are two different types of decoctions. These are strong and dried decoction (Nafiu et al. 2017). Strong decoctions are prepared by boiling plant parts in water for a longer period of time. Duration of boiling depends on the size of plant parts being boiled. Thereafter, the mixture is strained to separate the plant fragments from the liquid. The liquid may be consumed to treat gastro-intestinal ailments and internal parasites (Sanhokwe et al. 2016). For dried decoctions, the syrup obtained after boiling over a long period of

**Fig. 2** Traditional methods used for preparing and applying herbal medicines to animals

time is dried to form a powder. The powder is administered orally or topically on wounds. Decoctions are used to prepare herbal remedies of plants such as *Artocarpus heterophyllus*, *Morinda citrifolia*, *Hibiscus rosa-sinensis* (Panda and Mishra 2016), *Psidium guajava*, *Syzygium cumini*, *Andrographis paniculata* (Parthiban et al. 2016) and *Khaya senegalensis* (Noudèkè et al. 2017).

Efficacy of medicinal plant extracts

Plant species have different bioactive compounds that are extracted using different standard procedures (Sasidharan et al. 2010). Most experiments on efficacy of ethno-veterinary medicine are based on these phytochemicals. The compounds are relatively complex mixtures of metabolites in liquid or semi-solid state or in dry powder form (after removing the solvent) and are intended for oral or external use. Some examples of these compounds are alkaloids, flavonoids, phenols, reducing sugars, saponins, tannins, terpenoids, lectins, anthocyanins (Mujeeb et al. 2014; Hamadani et al. 2018). Water, ethanol, methanol, chloroform, ether and acetone are solvents commonly used for extraction of bioactive compounds in plants (Huie 2002; Zygmunt and Namiesnik 2003).

Mosquito larvicidal efficacy of plant extracts

According to Sukumar et al. (1991), there are about 344 plant species that exhibit mosquitocidal activity. *Aedes aegypti* mosquito is a vector of several diseases in livestock such as anthrax, Rift Valley fever and lumpy skin (Mukandiwa et al. 2015). Dichloromethane extract of *Zingiber purpureum* rhizome (Bandara et al. 2005), hexane leaf extracts of *Lantana camara* (Kumar et al. 2014) and *Clausena anisata* (Mukandiwa et al. 2015) and methanol extracts of *Leucas aspera* (Elumalai et al. 2017) exhibit larvicidal activity against *A. aegypti*. Similarly, Japanese encephalitis virus, endemic to the Asia-Pacific region, is primarily transmitted by the *Culex tritaeniorhynchus* mosquito causing reproductive losses and encephalitis in cattle, pigs and horses (Liu et al. 2018). Bark methanol extract of *Annona squamosa*, leaf methanol extract of *Chrysanthemum indicum* and leaf ethyl acetate extract of *Tridax procumbens* are effective against the larvae of *C. tritaeniorhynchus* (Kamaraj et al. 2011).

Acaricidal efficacy of plant extracts

In a study to identify alternative acaricides to control ticks in cattle, extracts of *Lantana camara* at 40% concentration had an efficacy of 57% (Moyo et al. 2009). On the same note, *Lippia javanica* aqueous leaf extracts at 10% and 20% w/v were effective at controlling cattle ticks but not as good as convectional acaricides (Madzimure et al. 2011). However, they can provide an effective tick control option for the

indigent populace in developing countries. Table 2 shows some of the acaricidal properties and efficacy of medicinal plants from different regions of the world. Most of these plant species are also found in the SPIC, hence may have similar effects in controlling parasites.

Antimicrobial efficacy of plant extracts

Common diseases in livestock are caused by bacteria species. Different plants have been reported to possess antimicrobial properties. In a study by Pawar et al. (2005), 50% *Aloe vera* gel concentrate inhibited growth of *Staphylococcus aureus*. *Aloe vera* extracts were also reported to inhibit growth of *Salmonella Typhimurium* and *Escherichia coli* (Redda et al. 2014). Leaf extracts of *Merremia emarginata* have shown better activity against *Bacillus cereus*, *Staphylococcus aureus*, *Pseudomonas aeruginosa* and *E. coli*. Other plant species proven to have antimicrobial properties are *Calophyllum inophyllum* (Adewuyi et al. 2014), *Cassia fistula* (Bhalodia and Shukla 2011), *Lumnitzera littorea* (Saad et al. 2011), *Acacia nilotica*, *Syzygium aromaticum* and *Cinnamum zeylanicum* (Khan et al. 2009). Similar plant species exhibiting antimicrobial properties can be used to address livestock health problems in SPIC.

Threats to traditional ethno-veterinary knowledge

Transfer of ethno-veterinary knowledge from one generation to another is either oral or based on apprenticeship, which implies that there is high risk of erosion if left undocumented (Yineger et al. 2008). Among the major factors that pose a threat to ethno-veterinary knowledge systems are acculturation, rural-urban migration, secrecy associated with medicinal plant knowledge, habitat destruction of medicinal plant resources, poor adoption of ethno-veterinary medicines by trained veterinary personnel and lack of pharmacodynamics information on plants and their potential applicability to a wide range of animal species.

Cultural change through modernization and religious transformation, formal education and commercial occupation transforms local knowledge systems. These cultural changes bring about alienation from nature and natural processes with the result that people often disregard the local natural environment (Saynes-Vásquez et al. 2013). Migration of people in search of education and job opportunities to urban centers often results in the loss of indigenous knowledge systems including ethnobotany and agrobiodiversity (Grau and Aide 2007). The custodians of traditional knowledge in most of the indigenous communities are old and dying while lack of interest among the younger generation has been reported from several countries around the world such as Nepal (Poudel and Singh 2016), Africa (Kayombo et al. 2013), India (Muthu et al. 2006; Panghal et al. 2010) and China (Shen et al.

Table 2 Acaricidal properties of different plant species from different parts of the world [source: Adenubi et al. 2016]

Plant species	Test results
<i>Aegle marmelos</i> (Linn.)	3 mg/ml and 2 mg/ml MeOH leaf extracts caused 100% acaricidal MR for <i>H. bispinosa</i> and 100% larvicidal MR for R.(B.) microplus at 24 h PT, respectively.
<i>Allium sativum</i> L.	100 mg/ml MeOH clover extracts caused 69% larvicidal MR, 85.83% IO, 100% failure of eclosion of eggs and 80% acaricidal MR within 15 days.
<i>Ananas comosus</i> L. Merr	500 mg/ml distilled water skin extracts caused 39.1% IO, 33.3% EHI, efficacy percentage of 59.4% and 0% larvicidal MR at 24 h PT.
<i>Azadirachta indica</i> A.	8 mg/ml EtOH leaf extracts caused 80% acaricidal MR and 34.0 mg egg mass reduction at 5 h PT.
<i>Capsicum frutescens</i> L.	75 mg/ml EtOH flower extracts caused 85% MR at 48 h PT; 25 mg/ml caused 86.8%. There was 99.9% acaricide efficacy.
<i>Carica papaya</i> L.	100 mg/ml EO seed extract caused 82.2% larvicidal MR, 100% IO and eclosion of eggs, 93.33% acaricidal MR within 15 days.
<i>Citrus reticulata</i> Blanco	100 mg/ml EtOH flower extracts caused 90.77% larvicidal MR 1–2 h PT and 100% acaricidal MR 24 h PT.
<i>Citrus sinensis</i> L.	100 mg/ml EtOH flower extracts caused 98.59% larvicidal 1–2 h PT and 99% acaricidal MR 24 h PT.
<i>Citrus hystrix</i> DC	100 mg/ml EtOH flower extracts caused 90.29% larvicidal MR 1–2 h PT and 98% acaricidal MR 48 h PT.
<i>Cymbopogon citratus</i>	125 mg/ml EtOH leaf extracts caused 98.78% larvicidal MR 1–2 h PT and 100% acaricidal MR 5 days PT.
<i>Datura stramonium</i> L	100 mg/ml MeOH leaf extracts caused 73.33% acaricidal MR at 15 day PT, 71.8% larvicidal MR, 77.17% IO and eclosion of eggs.
<i>Gynandropsis gynandra</i>	EO seed extracts at 0.1 µl conc. There was 98.9% repellency.
<i>Jatropha curcas</i> L	50 mg/ml EtOH leaf extracts caused 90% EHI at 30 days PT.
<i>Leucaena leucocephala</i>	4.8 mg/ml distilled water leaf extracts caused 66.79% larval MR at 48 h PT, 33.14% EHI and 1.8% ELI at 21 days PT.
<i>Ocimum basilicum</i> L	60 mg/ml, 80 mg/ml and 100 mg/ml crude CH extracts produced 70%, 80% and 100% acaricidal MR, respectively.
<i>Tagetes erecta</i> L	3 mg/ml and 2 mg/ml MeOH extract caused 70% acaricidal MR for <i>H. bispinosa</i> and 77% larvicidal MR for R.(B.) microplus 24 h PT.
<i>Tamarindus indica</i> L	500 mg/ml EtOH flower extracts caused 99% acaricidal MR 7 days PT.

2010). Liu et al. (2009) have reported that low incomes of traditional healers and a preference for high paying jobs are responsible for a lack of enthusiasm among youth to inherit oral traditions related to Tibetan medicine from the village elders in Xizang, Tibet. Furthermore, the secrecy associated with medicinal plant knowledge is a common phenomenon among indigenous communities. This arises through a desire to retain incomes for treatment, beliefs concerning loss of the potency of plants upon disclosure or inviting the wrath of deities (Tuasha et al. 2018; Upriety et al. 2012). It is also a common practice to transfer the knowledge only to the eldest son in the family. Such practices prevent free sharing of knowledge (Lulekal et al. 2014).

Ethno-medicinal knowledge has a high dependence on sustained supply of medicinal plant resources. Forests are an important habitat for medicinal plants. However, anthropogenic factors such as deforestation, forest degradation, habitat loss, overharvesting and overgrazing threaten medicinal plant resources in the wild (Lulekal et al. 2014). Among the Maasai community in Kenya, medicinal plant resources are declining due to deforestation for agriculture (Kiringe 2005) and over exploitation of forests for charcoal production and fencing (Kiringe and Okello 2005). In the Amazon, decreasing availability of the oils and bark of medicinal tree species has been

reported due to harvesting of the species for timber (Shanley and Luz 2003). Habitat fragmentation and biodiversity loss due to deforestation and forest degradation have also decreased the availability of medicinal plant resources in Fiji (Shah and Bhat 2019).

Integrating ethno-veterinary medicine in modern livestock health care systems

The use of ethno-veterinary medicine is now widely recognised, especially with majority of small-scale farmers living below the poverty datum line. Information on different medicines is restricted to community level and this varies from one region to another depending on the availability of plant resources. To date, the majority of the conventional medicines used to treat animals are synthetic and very few are from plant extracts. Drugs processed synthetically have problems pertaining to environmental contamination and leaving residues in food (Shankar et al. 2010). Furthermore, there is a growing concern on most of the synthetic drugs losing their potency. For example, excessive use of antibiotics as growth promoters in the poultry industry is believed to be linked to growing bacterial resistance to antibiotics in human medicine. Unlike the synthetic conventional drugs, ethno-veterinary

medicines are often safe to use (Phondani et al. 2010). However, there is a strong need to carry out rigorous testing of ethno-veterinary remedies before recommendation for utilization in treating animal ailments. This would ensure that deleterious effects on animals through ethno-medicine are minimised. Efforts are ongoing to ensure that some of the ethno-veterinary medicines become commercially available to farmers worldwide. Moringa products in some parts of Africa and India are now available in both informal and formal markets, where they are mainly sold in powder form for human consumption as an immune booster, but can also be used as medicine for animals (Eze et al. 2013). Other herbal medicines available in formal markets have antimicrobial properties and fly and/or insect repellents. *Aloe vera* plant extract, for example, is now widely used by veterinarians as an antibiotic. The product is available in the formal market in the form of gels, lotions and tablets. *Aloe vera* products can be used for treating various ailments ranging from wounds, ailment of the digestive system and boosting immune system (Gebremedhin 2018). Geraniol, a natural active ingredient extracted from different plants like *Coriandrum sativum*, *Cymbopogon martini*, *Geranium* spp., *Laurus nobilis*, *Rosa* spp. and *Thymus vulgaris*, is used as an insecticide (repellent) against external parasites in livestock (Chen and Viljoen 2010). Most of these medicines are, however, not available in either the formal or informal market sectors in the SPIC. There is a need to disseminate information to island inhabitants on the use of herbal medicine to treat livestock diseases.

Conclusion

This review indicates that herbal medicine may be used in place of conventional medicine to treat animal diseases in the SPIC. Herbal medicine is considered sustainable and adaptable to the rural farming communities because of its ease of availability, and simple modes of preparation and administration to animals. However, despite the abundance of bio-resources, not much research has been conducted on the use of plants for treating animal diseases in the SPIC. We believe that the present review will encourage future ethno-veterinary research among livestock disease management practices in the region. The documentation, conservation and awareness of herbal medicine with potential to treat animal diseases is highly recommended as a simple and cost effective strategy to address key production problems in the livestock sector.

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